

Heap extract-max Margin Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/heap-extract-max

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Abstract

`heap-extract-max` selects a unique maximizer. Margin $\gamma(s) > 2\varepsilon$ is necessary and sufficient for unique-max invariance under $\|\delta\|_\infty \leq \varepsilon$. Lean `LEAN_FULL` (reduction to `Argmax.Margin`).

This theorem is: Reduction.

1 Problem

The `heap-extract-max` operator returns the unique maximizer index i^* of scores $s \in \mathbb{R}^m$ ($m \geq 2$), up to the operator's definitional presentation (heap top, tournament winner, masked max, etc.).

2 Stability notion

Perturbations: $\|\delta\|_\infty \leq \varepsilon$. Let $\gamma(s) = s_{i^*} - \max_{j \neq i^*} s_j$. Stability: $\gamma(s) > 2\varepsilon$ implies i^* remains the unique maximizer of $s + \delta$. Sharpness: $\gamma(s) \leq 2\varepsilon$ admits an adversary destroying unique maximality.

3 Definitions

Definition 1 (Unique maximizer). i^* uniquely maximizes s if $s_j \leq s_{i^*}$ for all j and $s_j = s_{i^*} \Rightarrow j = i^*$.

Definition 2 (Margin). $\gamma(s) = s_{i^*} - \max_{j \neq i^*} s_j$.

4 Theorem

Theorem 3 (Margin invariance). If i^* uniquely maximizes s and $s_{i^*} - s_j > 2\varepsilon$ for all $j \neq i^*$, then for every $\|\delta\|_\infty \leq \varepsilon$, i^* uniquely maximizes $s + \delta$.

Theorem 4 (Sharpness). If some $j \neq i^*$ has $s_{i^*} - s_j \leq 2\varepsilon$, an admissible δ destroys unique maximality.

5 Intuition

Depressing the winner by ε and elevating a rival by ε consumes 2ε of margin. Strict margin above 2ε leaves the winner unique; otherwise a two-point adversary forces a tie or loss.

6 Examples

Example 5. Scores $(7, 5, 4)$, $i^* = 0$, $\gamma = 2$. For $\varepsilon = 0.8$, $2\varepsilon = 1.6 < 2$, so the winner is stable. For $\varepsilon = 1$, $2\varepsilon = 2 \geq \gamma$; $\delta = (-1, +1, 0)$ yields a tie between the first two scores.

7 Proof

Let $s' = s + \delta$ with $\|\delta\|_\infty \leq \varepsilon$. For $j \neq i^*$,

$$s'_{i^*} - s'_j = (s_{i^*} - s_j) + (\delta_{i^*} - \delta_j) \geq (s_{i^*} - s_j) - 2\varepsilon > 0,$$

using $s_{i^*} - s_j > 2\varepsilon$. Thus $s'_j < s'_{i^*}$ for all $j \neq i^*$, so i^* uniquely maximizes s' (Theorem 3).

For sharpness, pick j with $s_{i^*} - s_j \leq 2\varepsilon$, set $\delta_{i^*} = -\varepsilon$, $\delta_j = +\varepsilon$, and 0 elsewhere. Then $\|\delta\|_\infty \leq \varepsilon$ and $s'_{i^*} - s'_j = \gamma_{ij} - 2\varepsilon \leq 0$, so unique maximality fails (Theorem 4).

8 Formal statement

Lean module: `Research.Operators.HeapExtractMax.Margin`. Source file: `lean/Research/Operators/HeapExtractMax.Margin`. Theorems: `heap_extract_max_margin_invariance`, `heap_extract_max_margin_sharpness`. Definitional reduction of `Research.Operators.Argmax.Margin`. Certificate: `lean/certificates/heap-extract-max-margin`. Status: LEAN_FULL on Mathlib $\mathbb{R}$ (REAL_MATHLIB).

9 Proof dependencies

- `Research.Operators.Argmax.Margin` (`margin_invariance`, `margin_sharpness`).
- ℓ_∞ gap shrinkage by at most 2ε .
- No other operator theorems required.

10 Consequences

Any unique-max presentation of `heap-extract-max` inherits `Argmax` margin stability. Related: `argmax`, `tournament-winner`, `greedy-maximum-selection`.